

Indications and Selection of Patients for ECMO

In this chapter, we tackle one of the most sobering and important topics related to extracorporeal membrane oxygenation (ECMO) – selection of patients for ECMO. In doing so, we are trying to reconcile two very important and weighty decisions:

1. Can I choose someone for ECMO who would be likely to do well with ECMO?
2. Can I spare someone the risks of ECMO if they are unlikely to benefit?

These can be challenging to parse out and can weigh heavily on clinicians. To better understand this dynamic, let's return to our graph comparing ECMO to conventional care ([Fig. 7.1](#)).

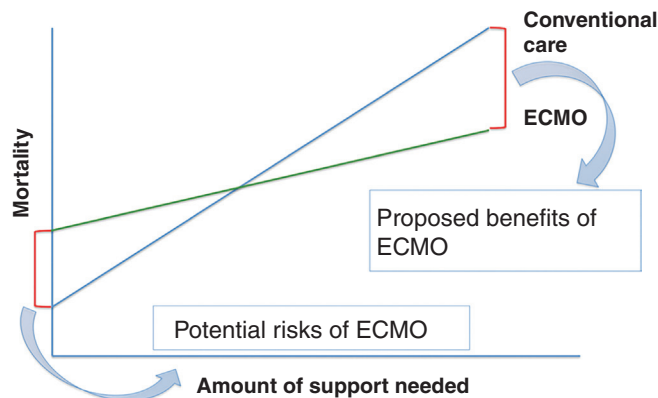


FIG. 7.1 Proposed risks and benefits of ECMO versus conventional care

The green line (ECMO) starts out the graph significantly higher than the blue line (conventional care), which represents that ECMO has a cost in terms of risk. Simply cannulating means inserting large cannulas into major blood vessels leading to all of the associated risks – pseudoaneurisms, damage to underlying structures, extremity ischemia, dissection, etc. Even if your team is excellent at cannulating with very few complications, you are exposing blood to foreign surfaces and altering the coagulation cascade, which could alter bleeding and hemodynamics down the road. Even from a resource standpoint, putting someone on ECMO can be expensive and resource intensive.

Given all this, you may be asking yourself some hard questions.

“Aren’t people either not sick enough to even consider ECMO or so sick that ECMO would not be of benefit?”

Or even more so, “Why even put someone on ECMO?”

These questions get to the heart of the challenge of selection for ECMO. While challenging, there is a sweet spot that exists – teams that are truly adept at selection are able to identify that select group of patients that is likely to do poorly without ECMO and will improve with ECMO (Fig. 7.2). This is the art of selection, which we will spend this entire chapter trying to refine.

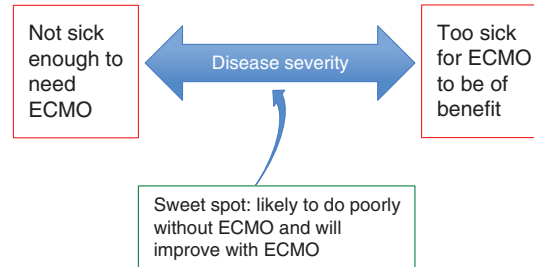


FIG. 7.2 The sweet spot for selection for ECMO

SO HOW CAN WE GET BETTER AT FINDING THIS SELECT GROUP OF PATIENTS?

To tease out this select patient population, let's consider the following rules, to help us hit that sweet spot with regards to selection for ECMO.

- Rule 1: **Reversible** etiology of cardiac or respiratory failure
- Rule 2: Conventional therapy is harming the patient more than helping
- Rule 3: Cardiac/respiratory failure has to be severe enough to justify the risks of ECMO
- Rule 4: Patient is not too far gone so that ECMO will not be of benefit

Let's explore a little more of what is behind each rule.

Rule 1: Reversible Etiology of Cardiac or Respiratory Failure

This is an essential rule and an acknowledgement that ECMO does little to fix the underlying disorder. Rather, a path must exist towards eventual recovery or destination (such as a transplant, long-term device, etc.).

Often, while recovery can exist, there is some barrier that is preventing this, whether it is the high ventilator requirements or escalating pressors. What the patient needs is time to allow for eventual recovery. This time is what ECMO can offer (Fig. 7.3).

The reversibility of the underlying cause of severe respiratory/cardiac failure will be a strong factor in a patient's candidacy for ECMO. Ideally, the more readily reversible the condition, the better, especially if the risk of decompensation without ECMO is significant. A patient with severe, refractory status asthmaticus, with an anticipated recovery of 24–48 hours on ECMO, has very different risk profile from a patient with ARDS from bacterial pneumonia, who may have an anticipated recovery of weeks to months. Likewise, a patient with a normal heart and cardiogenic shock from a pulmonary embolism may carry a very different risk profile from a patient with chronic cardiomyopathy and decompensated heart failure.

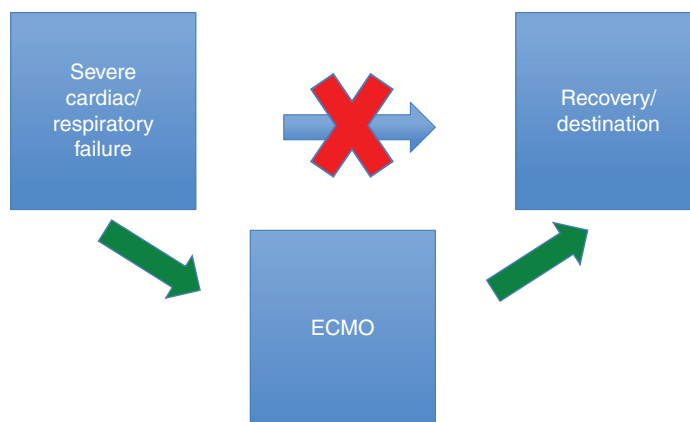


FIG. 7.3 The role of ECMO in arriving at recovery/destination in reversible conditions

Rule 2: Conventional Therapy Is Harming the Patient More Than Helping

We spent a great deal of time in Part 1 defining and outlining the dose-related toxicity of conventional care. All the things that can be done in the intensive care unit (ICU) to optimize oxygen delivery (pressors, inotropes, oxygen, positive pressure ventilation, fluid, blood) come with a cost that compounds with increasing dose.

These interventions combine diminishing returns with increased toxicity at higher levels. This means that the higher the dose of conventional supportive care is not only needed for sicker patients, but actually that conventional supportive care itself can be harmful at higher doses.

ECMO has to spare some degree of this toxicity. The more toxicity that can be potentially spared, the stronger the rationale for ECMO.

Rule 3: Cardiac/Respiratory Failure Has to Be Severe Enough to Justify the Risks of ECMO

It is one thing to say that conventional therapy is harming the patient more than helping. It is quite another to say that the degree that it is harming the patient is severe enough that sparing this harm is worth the risks of placing a patient on ECMO.

Let's suppose for example that you have a patient on a ventilator for pneumonia. The ventilator is set at 100% FiO₂, and he has been on this oxygen level for days.

Is there toxicity associated with this level of support? Certainly.

Does this toxicity justify the risks of ECMO? Maybe not.

To answer this question, we would need to really understand how dependent the patient was on this high level of support and what trajectory the patient was on. Otherwise, our graph may look something like Fig. 7.4, with the conventional support line increasing from left to right but not increasing to the degree that it would intersect with the ECMO line.

How can we quantify the response to the level of support and the trajectory? Let's consider a few parameters.

Respiratory Thresholds of Severity

Let's return to our patient with pneumonia. He may be on a harmful amount of oxygen with the ventilator set at 100%, but the true severity of his respiratory failure will be determined by his response to the high level of support. We can define this response by looking at the function of the lungs: to oxygenate (provide O₂) and to ventilate (remove CO₂). Remember that both of these functions can be performed, because these gases are able to flow down their relative concentration gradients at the level of the blood-alveolar interface (Fig. 7.5).

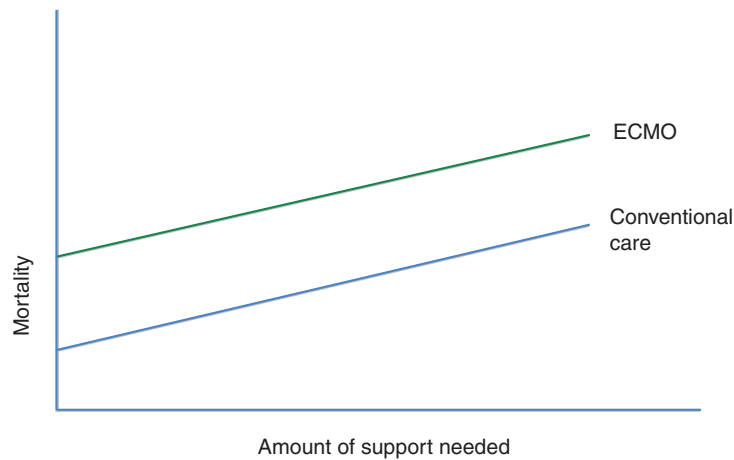


FIG. 7.4 The limited benefits of ECMO over conventional care in the case of lower disease severity

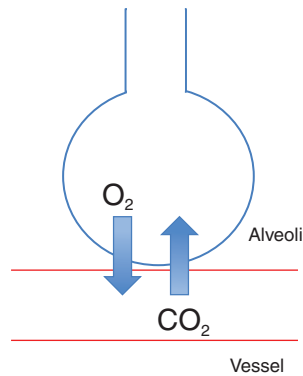


FIG. 7.5 The blood-alveolar interface

The oxygen that diffuses across the lungs is dissolved into the blood until it saturates the hemoglobin and can be efficiently delivered to the tissues. The amount of oxygen that is dissolved in the blood, or the PaO_2 , will give an indicator of how well the lungs are oxygenating – the worse the lungs are performing, the lower the PaO_2 .

A PaO_2 of 60 mmHg on room air (21% FiO_2) represents a very different clinical situation than a PaO_2 of 60 mmHg on 100% FiO_2 , as you would expect a much higher PaO_2 if you were administering 100% FiO_2 . The convention for quantifying this is to express the PaO_2 as a ratio to the FiO_2 , otherwise known as a $\text{PaO}_2:\text{FiO}_2$ ratio, or a P:F. As shunt increases, we have worsening P:F ratio that is less responsive to increases in FiO_2 . To combat this, we will have to increase alveolar recruitment through raising the mean airway pressure (Fig. 7.6).

Remember also that as lungs become progressively more compromised, compliance also decreases and it takes more pressure from the ventilator to obtain the same tidal volume with each breath (Fig. 7.7).

This can also contribute to a worsening ventilatory function of the lungs. When we are unable to adequately oxygenate/remove CO_2 despite optimal management of the ventilator, we can

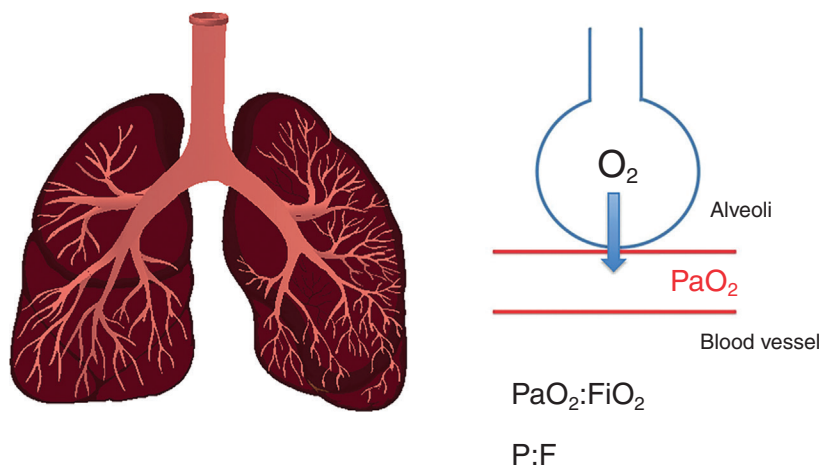


FIG. 7.6 The PaO_2 to FiO_2 ratio in respiratory failure. (Modified from YanGe_Tam/Shutterstock.com.)

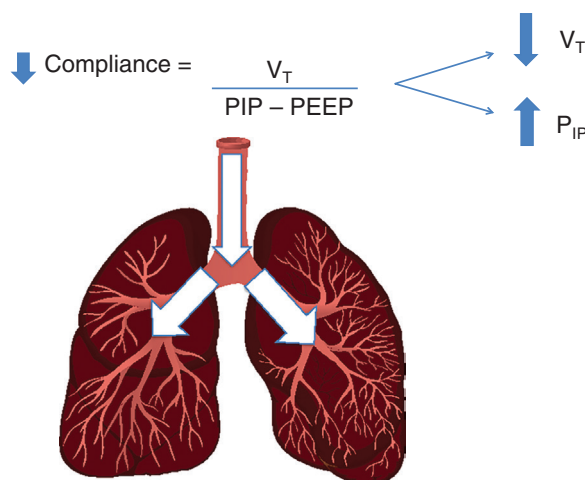


FIG. 7.7 The limitations of ventilation in the setting of decreased respiratory compliance. V_T , tidal volume, PIP , inspiratory pressure, PEEP , positive end expiratory pressure. (Modified from YanGe_Tam/Shutterstock.com.)

consider ECMO. This threshold is not well defined, but a reasonable starting point for discussion is a failure of oxygenation with $\text{P:F} < 100$ or a failure of ventilation with respiratory acidosis and a $\text{pH} < 7.25$ despite optimal mechanical ventilation.

Cardiac Thresholds of Severity

Cardiac thresholds for considering ECMO are not always as well defined, but they should be carefully considered.

In a similar fashion to our discussion of respiratory failure, the maintenance of cardiac output is essential for two tasks: maintenance of the vascular pressure head through generation of adequate mean arterial pressure and perfusion of end organs. Doing one without the other is insufficient for delivering oxygen and upkeep of metabolic function. Therefore, the evaluation for thresholds of cardiac failure should involve an evaluation of both the mean arterial pressure and the perfusion of end organs. Thus, let's examine the following parameters for consideration of VA ECMO (Fig. 7.8).

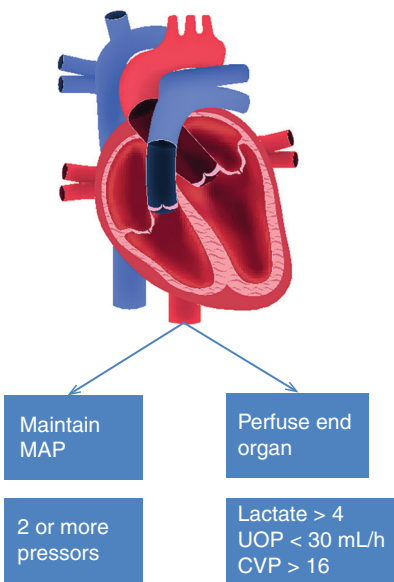


FIG. 7.8 Thresholds of cardiac failure in cardiogenic shock. (Modified from [Ody_Stocker/Shutterstock.com](#)). CVP, Central venous pressure; MAP, mean arterial pressure; UOP, Urine output.

Selection for VA ECMO should also include consideration of the physiology and nature of the insult and placed in the context of other forms of mechanical circulatory support (i.e., whether an intraaortic balloon pump/percutaneous ventricular assist device be a better form of support for the patient given their physiology). We will discuss this in much greater detail in the VA ECMO physiology sections.

Urgency for Consideration of ECMO

As a final note, these parameters should be put into the context of the clinical trajectory and urgency for consideration of ECMO. To return to our graphs, the intersection of the two lines would be considered the threshold for initiating ECMO (Fig. 7.9A). Therefore, in a situation where the patient is severely decompensated, like on the graph on the right, there would be a much lower threshold of consideration for ECMO (Fig. 7.9B).

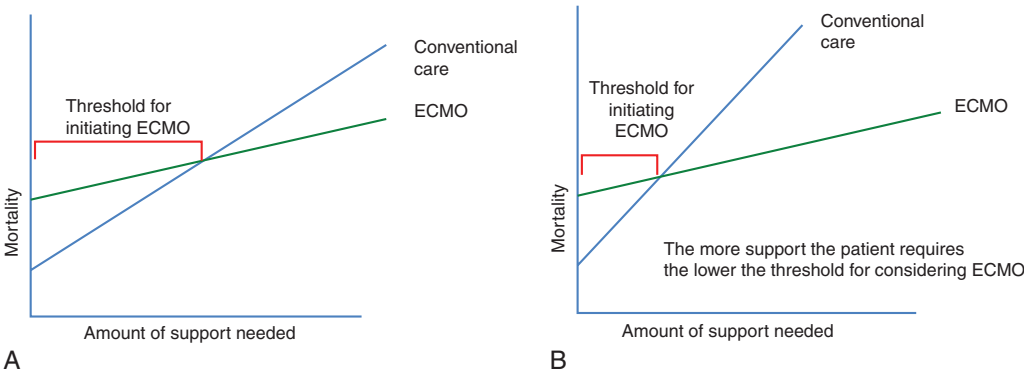


FIG. 7.9 Lower threshold for initiating ECMO in the case of higher disease severity

The worse the parameters, the more urgent the consideration for ECMO. For example, for two patients with respiratory failure with the following gases, Patient A might signal a failure to oxygenate/ventilate, with a more urgent consideration for ECMO:

Patient A: 7.15, PaCO_2 75, PaO_2 40 on 100%

Patient B: 7.35, PaCO_2 50, PaO_2 85 on 100% FiO_2

Rule 4: Patient Is Not Too Far Gone So That ECMO Will Not Be of Benefit

If you are going to offer ECMO, you want to make sure that it is of benefit. You do not want to offer it to someone who does not have a reversible condition and may never get better, nor would you want to offer it to someone who does not have some toxicity that you can spare them. Likewise, you would not want to offer it to someone who is not sick enough to justify the risks of initiating support.

In a similar fashion, you must carefully evaluate patients, especially sicker patients, for being so decompensated that ECMO will not be beneficial. This can include patients in multi-organ failure, patients with prolonged CPR, prolonged periods on mechanical ventilation, increasing age/comorbidities, and irreversible malignancies with a limited lifespan.

Sometimes, it can be relatively straightforward to deduce that initiation of ECMO is not worth the risks to the patient. Sometimes, it can be difficult to pick the time when the patient is in a nonrecoverable state, especially as you consider the case of escalating disease severity.

As illustrated in Fig. 7.10, the more decompensated a patient with cardiac/respiratory failure, the more likely they will benefit from ECMO, until a point when there is evidence of irreversible damage, and then the benefit rapidly becomes less apparent.

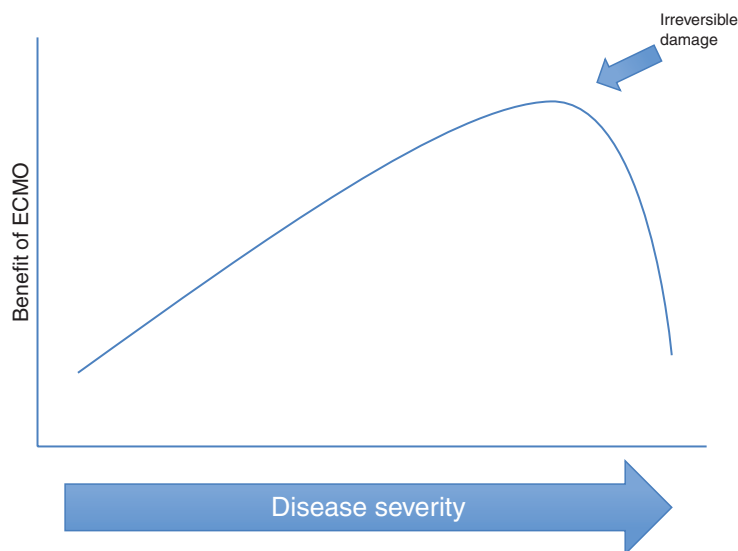


FIG. 7.10 Disease severity and the potential benefit of ECMO

What defines “irreversible damage”? This can be a difficult point to recognize. Is it a lactate cutoff in cardiogenic shock? Is it a specific time on the ventilator? You will find that the longer you evaluate patients for ECMO, the more nuanced of a sense you will develop for which patients will do well and which will do poorly (Fig. 7.11A).

To finish our analogy, patients too decompensated are like the graph on the right, where the slope of the conventional care line is so steep that it does not even intersect with the ECMO line (Fig. 7.11B).

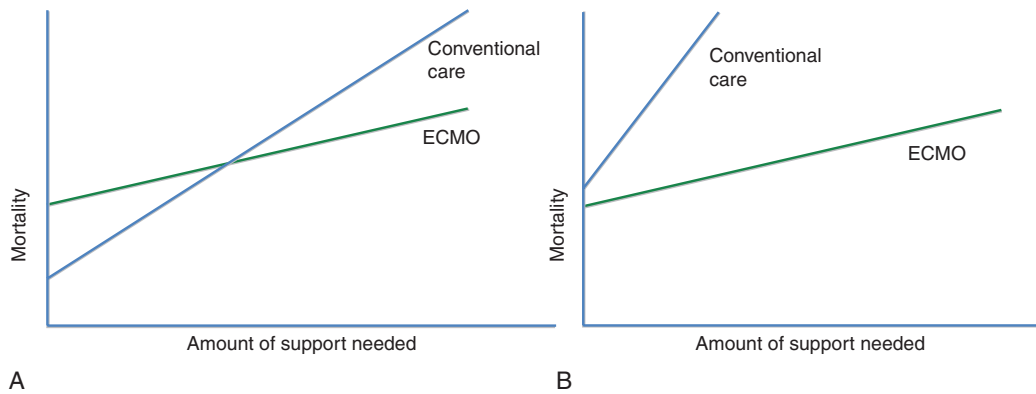


FIG. 7.11 Limited benefit of ECMO in the setting of irreversible damage

PUTTING IT ALL TOGETHER

When it comes to selecting for ECMO, I think intent is paramount. The four rules presented in this chapter are to help define that important population of patients that would benefit from ECMO who would likely do poorly without ECMO. They are not to make numbers and mortality look better. When we are considering whether we should initiate support, we need to define and quantify the gap between outcomes with and without ECMO and weigh that gap against the ability to provide that care. A large ECMO program performing its 200th ECMO run may be more willing to take on a more complex patient than a program with less experience, resources, and capabilities.

At the end of the day, selecting patients for ECMO requires great humility and a realization of the limits of our knowledge and predicting capability. Although we may be not able to predict the overall outcome with absolute certainty, if we are approaching every evaluation with the desire to find patients who are likely to derive benefit while eliminating the consideration of patients who are likely to derive harm, we are certainly on the right path.

SUGGESTED READING

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